

Hybrid parameters.

Hybrid parameters are four (4) constants which describe the behaviour of a two-port linear network. The four variables h_{11} , h_{12} , h_{21} , and h_{22} are called hybrid parameters.

For a given two-port network of Fig 1.0,

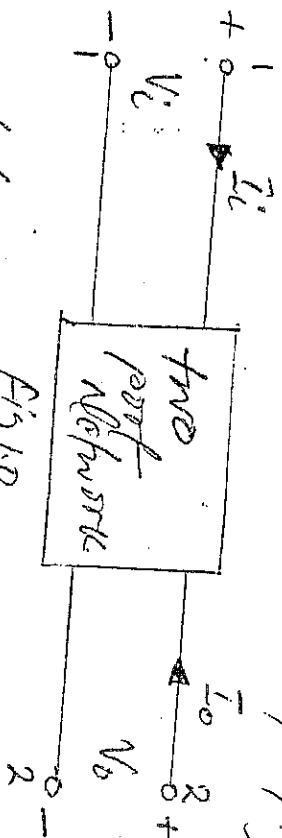


Fig 1.0

We can deduce:

$$V_i = h_{11} I_i + h_{12} V_o \quad (1)$$

$$I_o = h_{21} I_i + h_{22} V_o \quad (2)$$

$$\therefore h_{11} = h_i \text{ (input resistance)} = V_i / I_i \mid V_o = 0$$

$$h_{12} = h_r \text{ (reverse voltage gain)} = V_i / V_o \mid I_i = 0$$

$$h_{21} = h_f \text{ (forward current gain)} = I_o / I_i \mid V_o = 0$$

$$h_{22} = h_o \text{ (output admittance)} = I_o / V_o \mid I_i = 0$$

The hybrid equivalent circuit is given by:

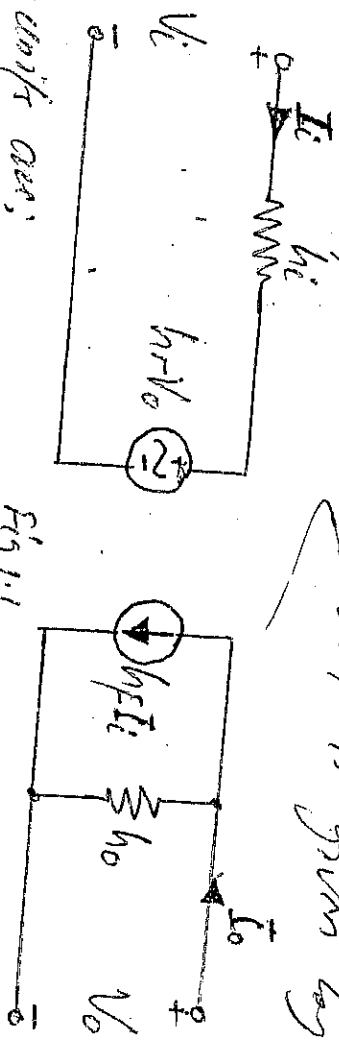


Fig 1.1

These units are:
 h_i (input resistance) = Ω , h_r (reverse voltage gain) = V_o unit.
 h_f (forward current gain) = A_0 unit, h_o (output admittance) = Siemens

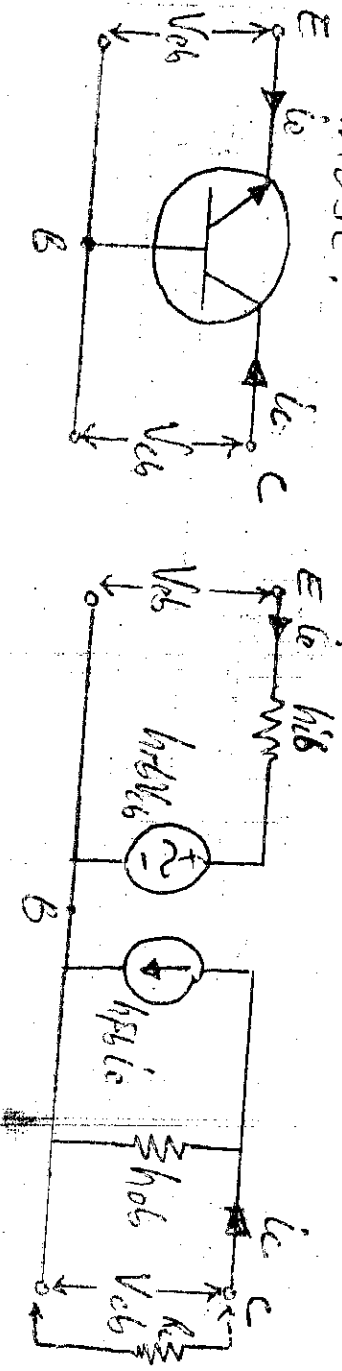
2 Hybrid Model of a Transistor in different mode.

A transistor is actually a three-terminal device. However any of the terminals is used as common with both output and input. Thus virtually it has two input and two output terminals.

The following diagrams show different transistor configurations and corresponding hybrid models.

1.2.1

CE - MODE:



Current gain $A_i = \frac{I_o}{I_i} = \frac{I_c}{I_b}$

Applying KCL to the output current gives,

$$\Rightarrow I_c = h_{FE} I_b + h_{OE} V_{CE}$$

$$I_o = I_c, \quad I_i = I_b$$

$$\text{Also } V_{CE} = -I_c R_L$$

$$I_c = h_{FE} I_b - I_c h_{OE} R_L$$

$$I_c (1 + h_{OE} R_L) = h_{FE} I_b$$

$$= \frac{I_c}{I_b} = \frac{h_{FE}}{1 + h_{OE} R_L}$$

$$A_i = h_{FE} \approx 1$$

But if $h_{OE} R_L \ll 1$

avg gain $A_v = V_o / V_i$ Applying KVL to the output circuit gives,

$$= I_c h_{OE} + h_{OE} V_{CE}$$

$$I_c = \frac{I_c (1 + h_{OE} R_L)}{h_{FE}}$$

$$I_c = - \frac{V_{CE}}{R_L}$$

$$\therefore i_c = - \frac{V_{cb} (1 + h_{ob} R_L)}{h_{fb} R_L}$$

$$\therefore V_{cb} = - \frac{V_{cb} h_{ib} (1 + h_{ob} R_L)}{h_{fb} R_L} + h_{rb} V_{cb}$$

$$V_{cb} h_{fb} R_L = - V_{cb} h_{ib} (1 + h_{ob} R_L) + h_{fb} R_L \cdot h_{rb} V_{cb}$$

$$V_{cb} h_{fb} R_L = - V_{cb} h_{ib} + V_{cb} h_{rb} h_{fb} R_L$$

$$V_{cb} h_{fb} R_L = - V_{cb} [h_{ib} + h_{rb} h_{fb} R_L]$$

$$\therefore A_v = \frac{V_{cb}}{V_{cb}} = - \frac{h_{fb} R_L}{h_{ib}} \quad \text{if } h_{rb} h_{fb} R_L \text{ is small.}$$

$$\text{Input impedance } Z_i = \frac{V_i}{I_i}, \quad V_i = V_{cb}, \quad I_i = i_c$$

From the input circuit;

$$V_{cb} = i_c h_{ib} + h_{rb} V_{cb} \quad \text{but } V_{cb} = - i_c R_L$$

$$V_{cb} = i_c h_{ib} + i_c h_{rb} R_L$$

$$\therefore Z_i = \frac{V_{cb}}{i_c} = h_{ib} - h_{rb} R_L$$

but in practice the factor $h_{rb} R_L$ is usually small.

$$Z_i \approx h_{ib}$$

$$\text{Output impedance } Z_o = V_o / i_o$$

From the output circuit.

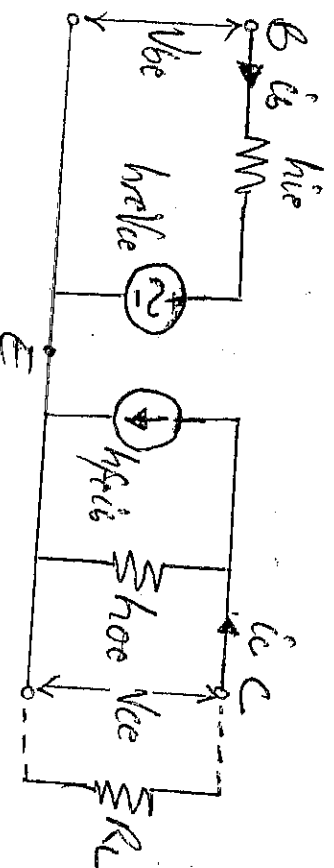
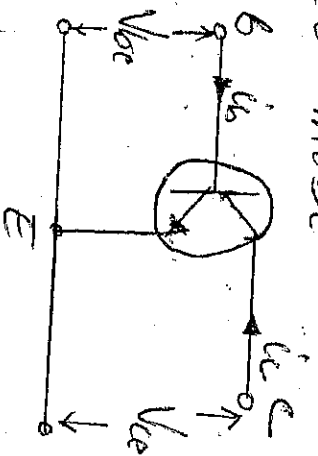
$$i_o = V_o h_o + h_{fb} i_c$$

$$i_c = i_o, \quad V_{cb} = V_o$$

$$Z_o = \frac{1}{h_o}$$

1.2.2

CE-MODE



Current gain $A_i = \frac{i_o}{i_i}$

Applying KCL to the output circuit gives;

$$i_o = h_{fe} i_b + h_{oe} V_{ce}, \quad \text{Also } V_{ce} = -i_c R_L$$

$$i_o = i_c = h_{fe} i_b - h_{oe} i_c R_L$$

$$i_c (1 + h_{oe} R_L) = h_{fe} i_b$$

$$\therefore A_i = \frac{i_c}{i_b} = \frac{h_{fe}}{1 + h_{oe} R_L}$$

$$A_i = h_{fe} \approx \beta \quad \text{if } h_{oe} R_L \ll 1$$

Voltage gain $A_v = V_o / V_i$

Applying KVL to the output circuit gives.

$$V_{be} = i_b h_{ie} + h_{re} V_{ce}$$

$$\text{but } i_b = \frac{i_c (1 + h_{oe} R_L)}{h_{fe}}, \quad i_c = -\frac{V_{ce}}{R_L}$$

$$i_b = \frac{-V_{ce}(1 + h_{oe}R_L)}{h_{fe}R_L}$$

$$\therefore V_{be} = \frac{-V_{ce} h_{ie}(1 + h_{oe}R_L)}{h_{fe}R_L} + h_{ie}V_{ce}$$

$$\text{Similarly } A_v = \frac{V_o}{V_i} = \frac{V_{ce}}{V_{be}} = \left(\frac{-h_{fe}R_L}{h_{ie}} \right)$$

$$\text{Input impedance } Z_i = \frac{V_i}{I_i}$$

$$\text{Similarly } Z_i = h_{ie}$$

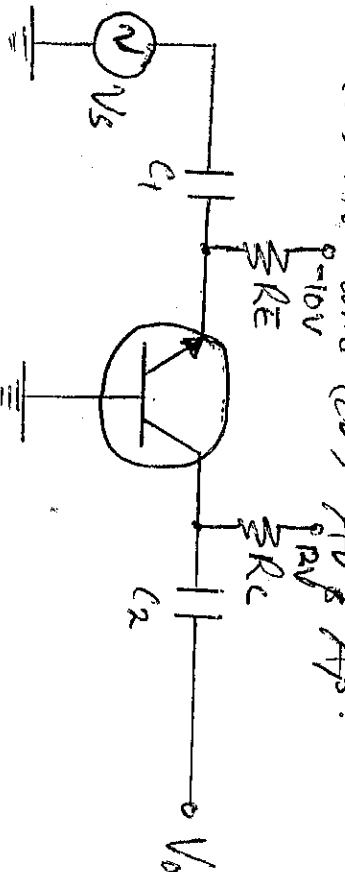
$$\text{Output impedance } Z_o = \frac{V_o}{I_o}$$

$$\text{Similarly } Z_o = \frac{1}{h_{oe}}$$

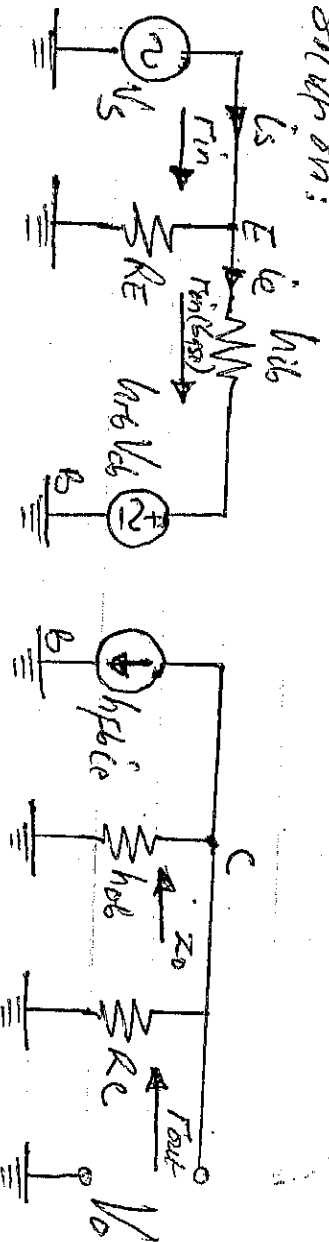
$$\text{Power gain } A_p = A_v \times A_i$$

1. Example: The common base (CB) transistor circuit shown below has the following set of parameters -

$h_{ib} = 36\Omega$, $h_{fe} = 0.98$, $h_{re} = 5 \times 10^{-4}$, $h_{oe} = 10^{-6} \text{ Siemens}$
 with $R_E = 5k\Omega$ and $R_C = 10k\Omega$. Calculate (i) r_{in}
 (ii) T_{out} (iii) A_i and (iv) A_v & A_p .



solution:



$$(i) r_{in}(b_{751}) = r_{ic} = h_{ib}$$

$$\therefore r_{in} = h_{ib} \parallel R_E$$

$$r_{in} = \frac{36 \times 5000}{36 + 5000} = 35.74 \Omega$$

$$\Rightarrow r_{in} \approx h_{ib} = 36 \Omega$$

$$(ii) r_{out} = r_{o} \parallel R_c$$

$$\text{where } Z_o = \frac{1}{h_{ob}} = \frac{1}{10^{-6} S} = 1 M\Omega$$

$$\therefore r_{out} = \frac{1000000 \times 10,000}{1000000 + 10,000} = 9900.99 \Omega$$

$$\therefore r_{out} \approx R_c = 10,000 \Omega$$

$$(iii) A_i = \frac{h_{fb}}{1 + h_{ob} R_L} \quad \text{where } R_L = R_c$$

$$A_i = \frac{0.98}{1 + (10^{-6} \times 10,000)} = \frac{0.98}{1 + 0.01}$$

$$A_i = 0.97029$$

$$\therefore A_i \approx h_{fb} = 0.98$$

$$(iv) A_v = \frac{h_{fe} R_L}{h_{ie}} \quad \text{where } R_L = R_C$$

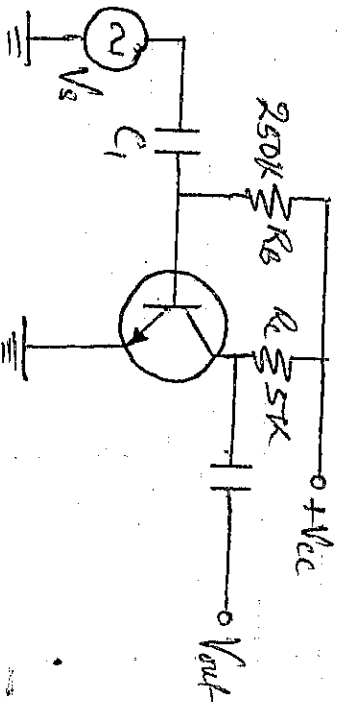
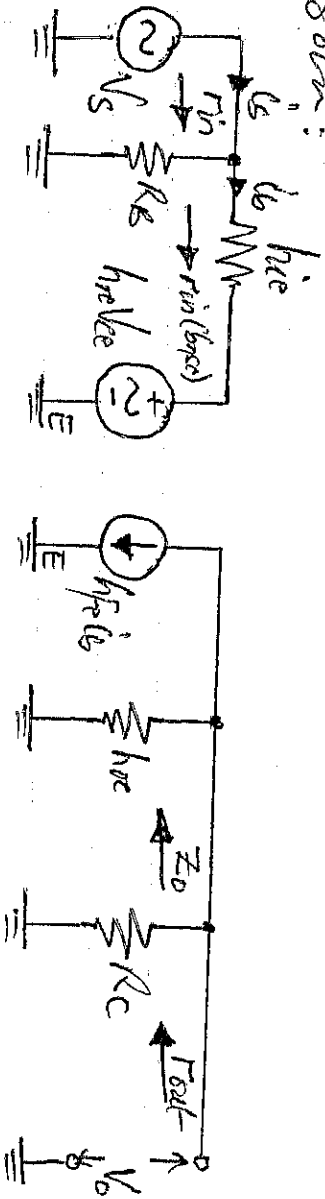
$$A_v = \frac{0.98 \times 10,000 \Omega}{36 \Omega} = 272.22$$

$$A_{180} A_p = A_v A_i = 272.22 \times 0.98$$

$$A_p = 266.78$$

2. Example: In the CE transistor circuit shown below, the transistor parameters are: $h_{ie} = 2k\Omega$, $h_{fe} = 100$, $h_{re} = 5 \times 10^{-4}$, $h_{oe} = 2 \times 10^{-5} S$. Calculate $A_{v_{in}}$, (ii) T_{out} (iii) A_i , (iv) A_v , (v) A_p with $R_B = 250k\Omega$, $R_C = 5k\Omega$. Use the approximate formulas given in 1.2.2.

Soln:



$$(i) r_{in}(base) = Z_i = h_{ie} = 2k\Omega$$

$$\therefore R_{in} = Z_i \parallel R_B = \frac{2 \times 250}{2 + 250}$$

$$R_{in} = 1.98k\Omega$$

$$(ii) r_{out} = Z_o \parallel R_c$$

$$\therefore Z_o = \frac{1}{h_{oe}} = \frac{1}{2 \times 10^{-5} S} = 50 k\Omega$$

$$r_{out} = \frac{50 \times 5}{50 + 5} = 4.55 k\Omega$$

$$(iii) A_i \approx h_{fe} = 100$$

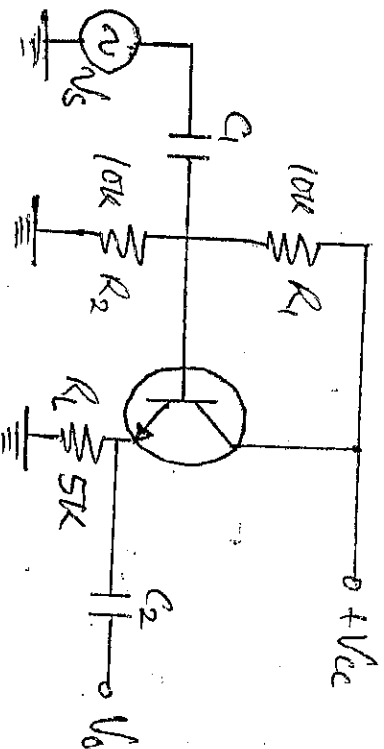
$$(iv) A_v = -\frac{h_{fe} R_L}{h_{ie}} \quad \text{where } R_L = R_c.$$

$$A_v = -\frac{100 \times 5000 \Omega}{2000 \Omega} = -250$$

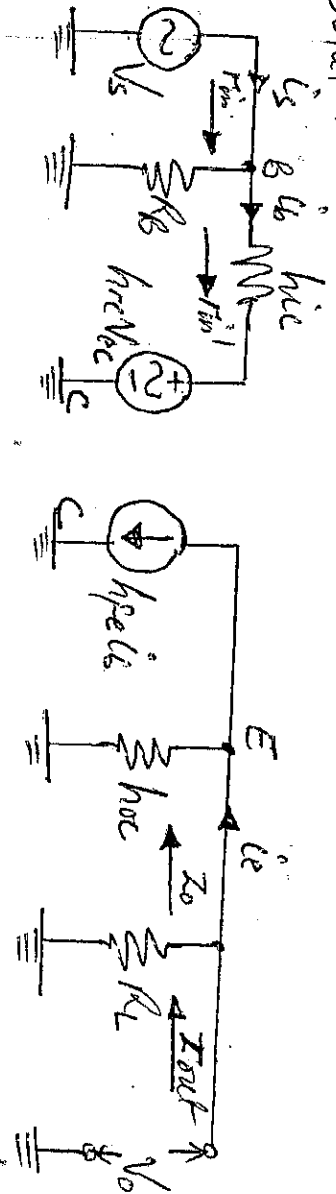
$$(v) A_p = A_i \times A_v = 100 \times 250$$

$$A_p = 25,000.$$

3. Example: Compute for the CC circuit with transistor parameters, $h_{ie} = 2k\Omega$ and $h_{fe} = 100$ the following.
(i) r_{in} , (ii) r_{out} , (iii) A_i , (iv) A_v .



Solution.



$$(i) V_{in} = i_b h_{ie} + h_{re} V_{ce}$$

$$\text{but } h_{re} \approx 1$$

$$V_{ce} = V_o = i_o R_L$$

$$\begin{aligned} \therefore V_{in} &= i_b h_{ie} + i_o R_L \\ &= i_b h_{ie} + h_{fe} i_b R_L \\ &= i_b (h_{ie} + h_{fe} R_L) \end{aligned}$$

$$\therefore r_{in}' = \frac{V_{in}}{i_b} = h_{ie} + h_{fe} R_L = Z_i$$

$$r_{in}' = 2000\Omega + 100 \times 5000 = 502\text{K}\Omega.$$

$$\therefore r_{in} = r_{in}' \parallel R_B \quad \text{where } R_B = R_1 \parallel R_2.$$

$$r_{in} = 502\text{K}\Omega \parallel 10\text{K}\Omega \parallel 10\text{K}\Omega = 4.95\text{K}\Omega$$

$$(ii) Z_o = \frac{h_{ie} + (R_1 \parallel R_2)}{h_{fe}} = \frac{2\text{K}\Omega + 5\text{K}\Omega}{100}$$

$$Z_o = 70\Omega.$$

$$r_{out} = Z_o \parallel R_L = \frac{70 \times 5000}{70 + 5000}\Omega$$

$$r_{out} = 69.03\Omega.$$

$$(iii) A_v = \frac{i_c}{i_b} \approx \frac{i_c}{i_b} \approx \frac{h_{fe} i_b}{i_b} \approx h_{fe} = 100$$

$$(iv) A_v \approx 1$$

2.1

THE THYRISTOR

The prominent member of the Thyristor family is the silicon controlled rectifier (SCR). The SCR is a three terminal semi-conductor switching device.

A SCR device can act as a true switch. It can change alternating current into direct current and at the same time can control the amount of power fed to a load. Thus SCR combines the features of a triac and a transistor.

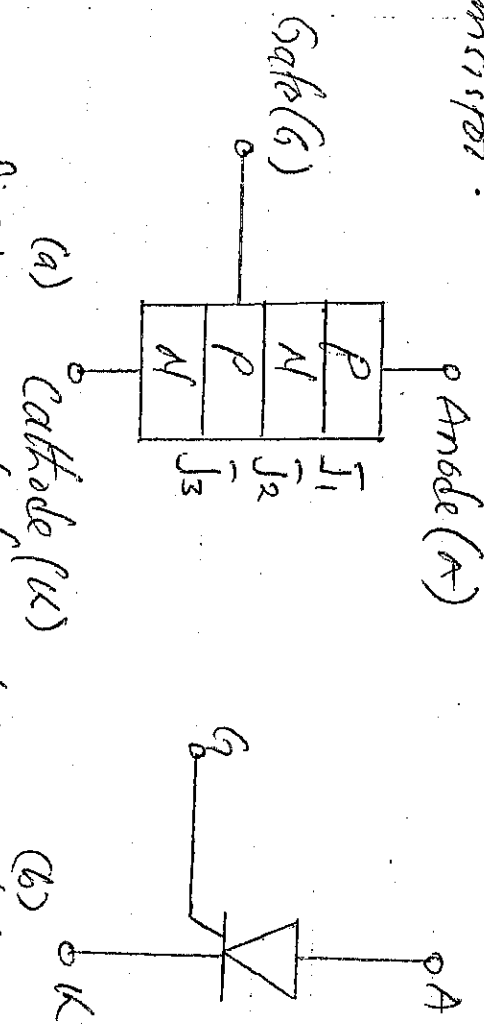


Fig 2.1: (a) construction and (b) symbol of SCR.

2.1.2 Operation of SCR:

Fig 2.2 shows the SCR circuit with the gate open. Under this condition, junction J_2 is reverse biased while junctions J_1 and J_3 are forward biased. With J_2 reverse biased, no current flows through the load R_L and the SCR is cut off. However, if the applied voltage is gradually increased, a stage is reached when reverse bias junction J_2 breaks down. The SCR now conducts heavily and is said to be "on state". The applied voltage at which SCR conducts heavily without gate voltage is called Breakover Voltage.

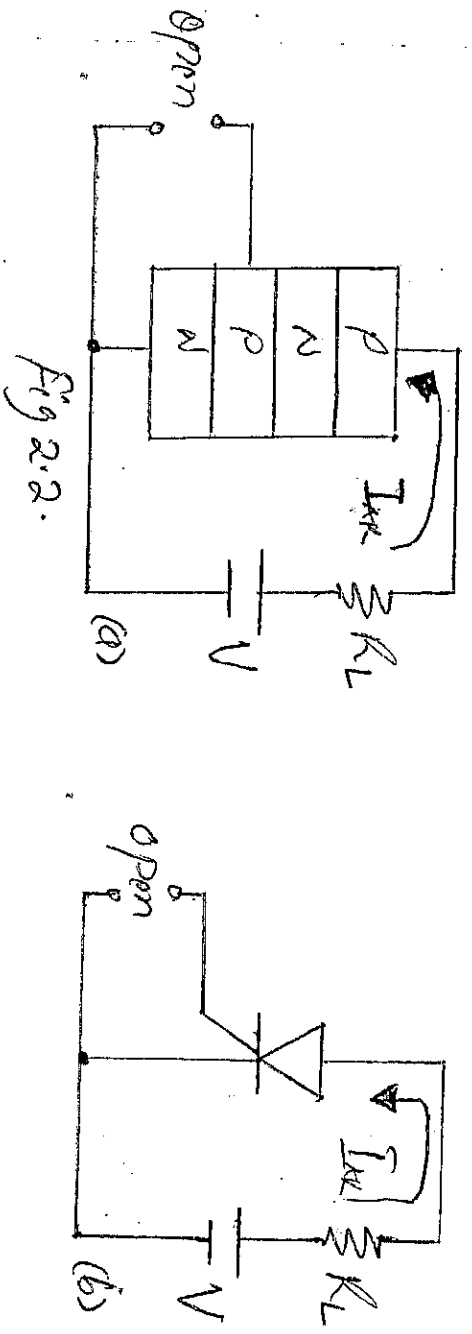
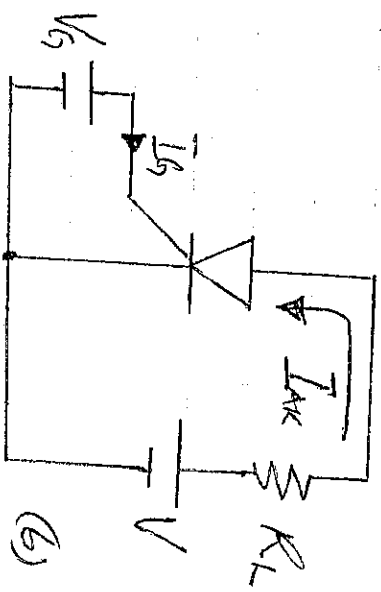
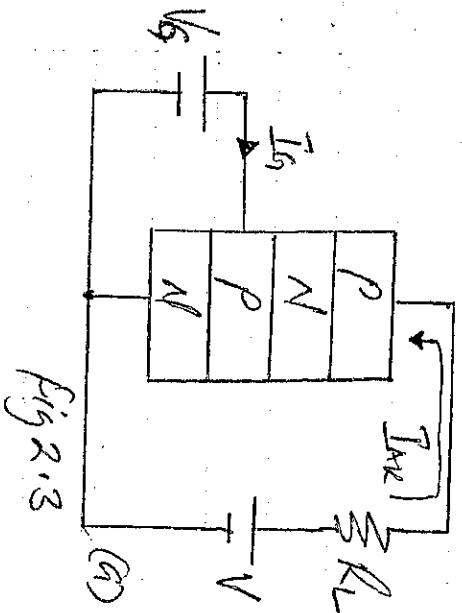


Fig 2.3 shows the SCR circuit when the gate is positive w.r.t. cathode. The SCR can be made to conduct heavily at smaller applied voltages by applying a small positive potential at the gate. Now junction J_3 is forward biased and junction J_2 is reversed biased. The electrons from n-type material start moving across junction J_3 toward the up, whereas holes from p-type start moving down. As soon as the gate current starts flowing, anode current increases. This process continues and in an extremely small time, junction J_2 breaks down and SCR starts conducting heavily. Once SCR starts conducting, the gate loses all control. Even if gate voltage is removed, the anode current does not decrease at all. The only way to stop conducting is to reduce the applied voltage to zero.



2.1.3 V-I Characteristics of SCR

(i) Forward Characteristics: When the anode is positive w.r.t. Cathode, the curve of V and I is called forward characteristics. In Fig 2.4, line OABC is the forward characteristic of SCR at $I_G = 0$. If the supply voltage is increased from zero, a point is reached (point A) when the SCR starts conducting. If proper gate current is made to flow, SCR can conduct at much smaller supply voltage.

(ii) Reverse Characteristics: When anode is negative w.r.t. Cathode, the curve PDE is called the reverse characteristic. When operated with a.c. supply, the reverse voltage does come across. If the reverse voltage is gradually increased, at first the anode current remains small (no leakage current) and at some point, avalanche breakdown occurs and the SCR starts conducting heavily in the reverse direction.

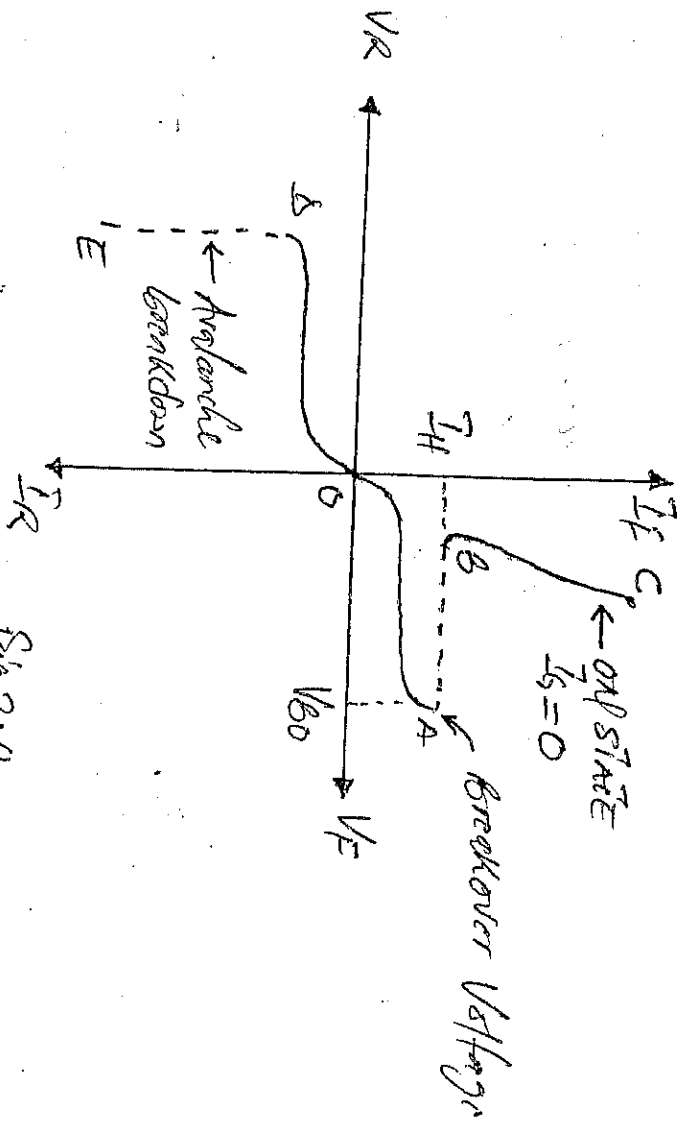
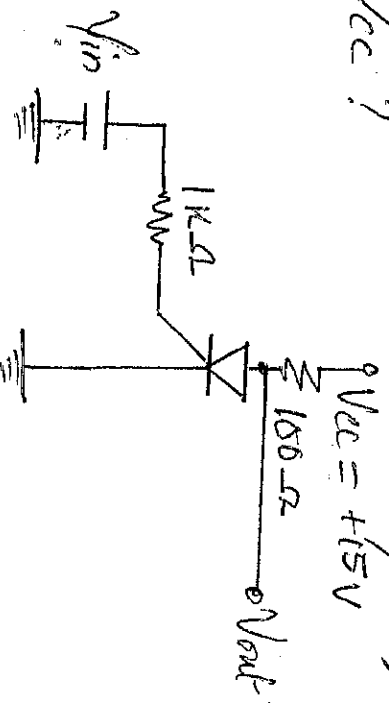


Fig 2.4

Example: A $1K\Omega$ resistor is connected in series with the gate of an SCR, the gate trigger voltage $V_g = 0.7V$.
 (i) What is the output voltage when the holding current $I_H = 6mA$?
 (ii) What is the input voltage that triggers the SCR?
 (iii) If V_{ec} is decreased until the SCR opens, what is the value of V_{ec} ?



Solution

i) When SCR is off, there is no current through the 100Ω resistor.

$$\therefore V_{out} = \text{Supply Voltage } V_{cc} = 15V$$

ii) Looking at the input circuit and using KVL:

$$V_{in} = V_T + I_T R = 0.7 + 7mA \times 1k\Omega$$

$$V_{in} = 7.7V$$

iii) In order to open the SCR, the V_{ce} must be reduced so that anode current is equal to I_H .

$$\therefore I_H = \frac{V_{cc} - V_T}{100\Omega}$$

$$\text{or } V_{cc} = I_H (100\Omega) + V_T$$

$$= 6mA \times 100\Omega + 0.7 = 1.3V.$$

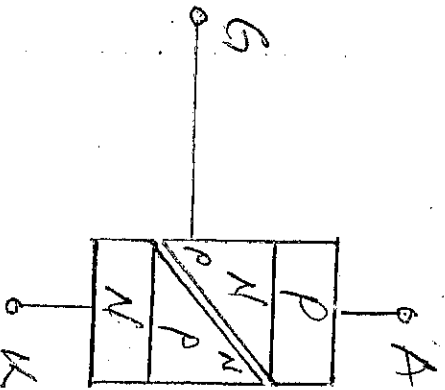
2.1.4

Applications of SCR

Main application of an SCR is as a power control device. Other common areas of its application include.

- 1) Relay controls
- 3) Static Switches
- 5) Inverters
- 7) Heater controls
- 2) Regulated power supply
- 4) Motor controls
- 6) Battery charger
- 8) Phase control

2.1.5

Transistor Analogy of SCR:

A

 I_{E1} Q₁ I_{B1} I_{E2} I_{C1} I_{E2} I_{B2} I_{C2} I_{E1} I_{C2} I_{E2} I_{C1} I_{E1} I_{C2} I_{E2

The basic operation of a SCR can be described by using the transistor analogy as shown in Fig 2.5. For this purpose, SCR is split into two 3-layer transistor structure as shown. As seen transistor Q_1 is a PNP transistor whereas Q_2 is an NPN transistor.

2.1.6 SCR Firing and Triggering:

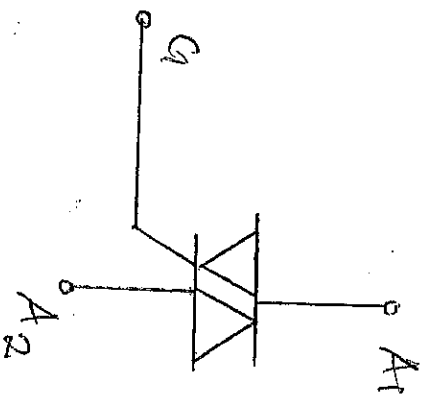
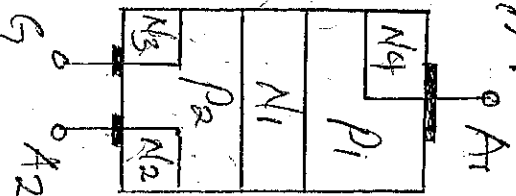
Usually, SCR is operated with an anode voltage slightly less than the forward breakover voltage V_{BO} and is triggered into conduction by a low power gate pulse. Once switched on, gate has no further control on the device current.

Gate signal used for triggering can be:

- (a) d.c. firing signal.
- (b) pulse signal.

2.2 TRIAC:

It is a 5-layer bi-directional device which can be triggered into conduction by both positive and negative voltages at its anodes and with both positive and negative triggering pulses at its gate. It behaves like two SCRs connected in parallel, upside down with respect to each other.



(a) Construction of Triac

(b) Schematic symbol.

Fig 2.6

2.2.1 V-I Characteristics:

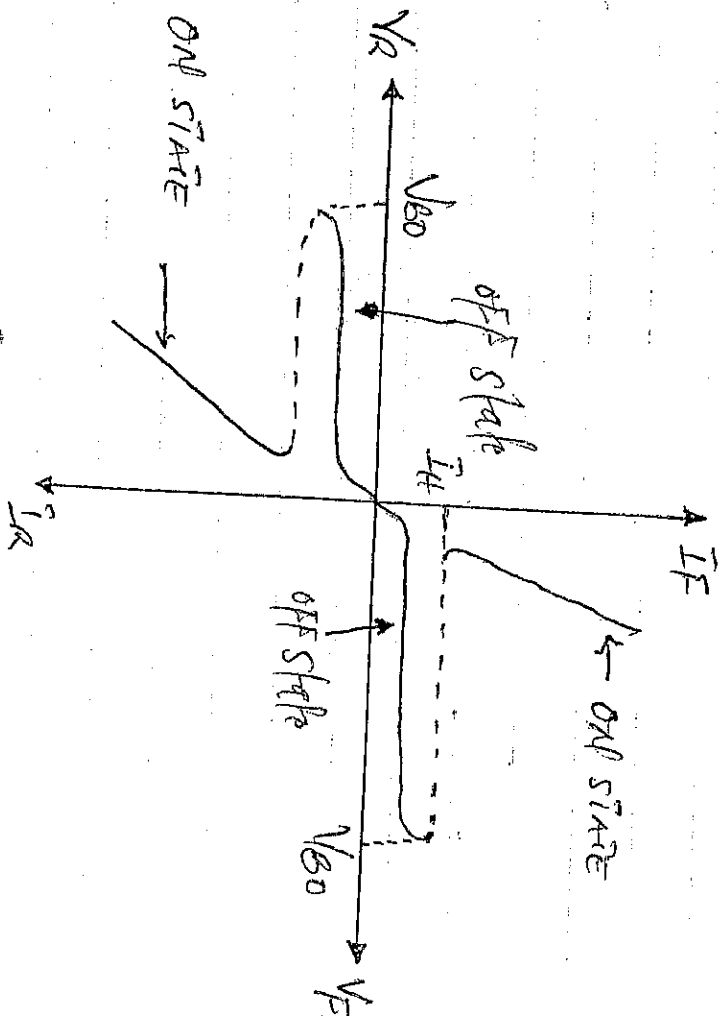


Fig 2.7.

2.2.2 Applications of Triac;

1. As static switch to turn ac power ON and OFF.
2. For Minimizing radio interference
3. For light control
4. For Motor speed control.

2.3 DIAC:

As shown in Fig 2.8, a diac is nothing but a triac with out its gate terminal. The operation of a diac can best be explained by imagining it as two diodes connected in series. Voltage applied across it in either direction turns on one diode, reverse-biasing the other. Hence, it can be switched from OFF to ON state for either polarity of the applied Voltage.

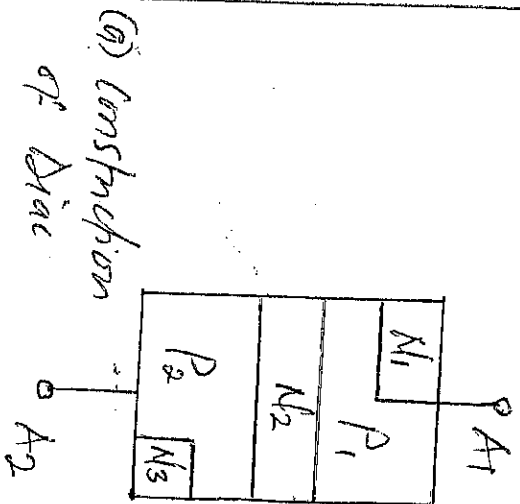
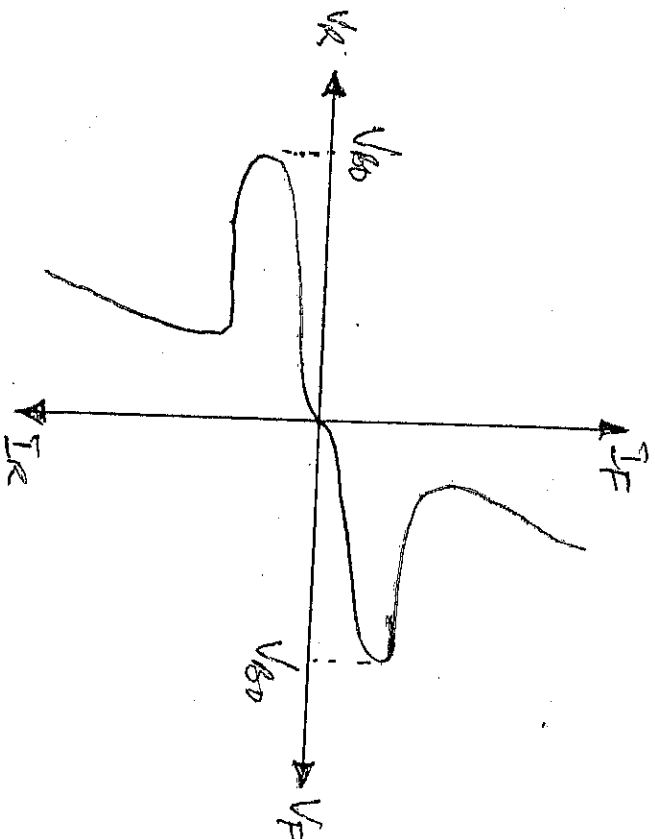
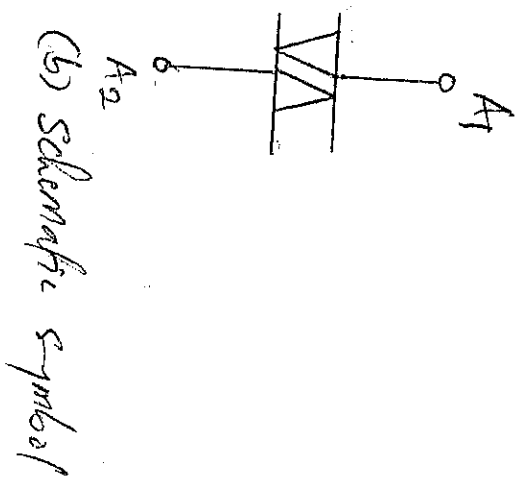


Fig 2.8:

Fig 2.9: $V-I$ characteristics of the Diac.

3.1 Applications of Diac:

1. In Counters, registers and timing circuits of Computers
2. Pulse generators
3. Voltage sensor
4. Oscillators

3.0 DIFFERENTIAL AMPLIFIERS.

Differential amplifier is a basic building block of an op-amp. The function of a differential amplifier is to amplify the difference between two input signals.

3.1 Operation of differential Amplifier circuit.

Let us consider two emitter-biased circuits as shown in fig 3.1(a). The two transistors Q_1 and Q_2 have identical characteristics i.e., $R_{E1} = R_{E2}$, $R_{C1} = R_{C2}$ and the Magnitude $+V_{CC}$ is equal to $-V_{EE}$. The V_o voltages are measured with respect to ground.

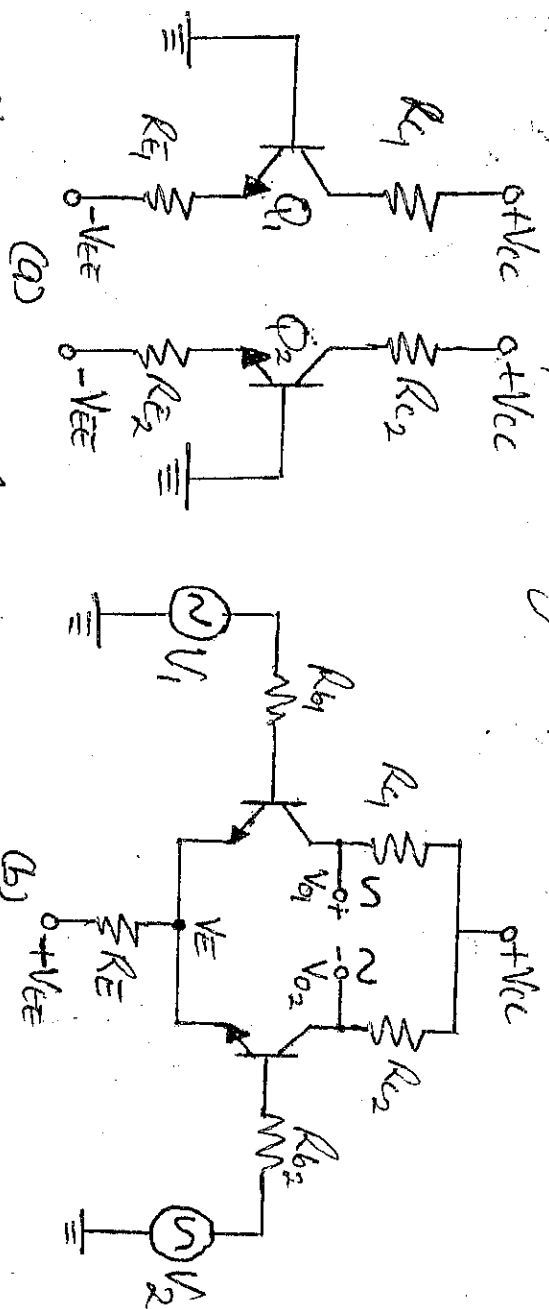


Fig 3.1: Differential Amplifier circuit.

To make a differential amplifier, the two circuits are connected as shown in fig 3.1(a). Ideally, the output voltage is zero when the two inputs are equal. When V_1 is greater than V_2 , the output voltage with the polarity shown appears. When V_1 is less than V_2 , the output voltage has the opposite polarity.

3.2 Differential Amplifiers configurations:

Differential amplifiers are of different configurations. The four (4) differential amplifier configurations are as

Follows:

- i) Dual input, balanced output differential amplifier.
- ii) Dual input, unbalanced output differential amplifier.
- iii) Single input, balanced output differential amplifier.
- iv) Single input, unbalanced output differential amplifier.

These configurations are shown in Fig 3.2 a, b, c, and d respectively.

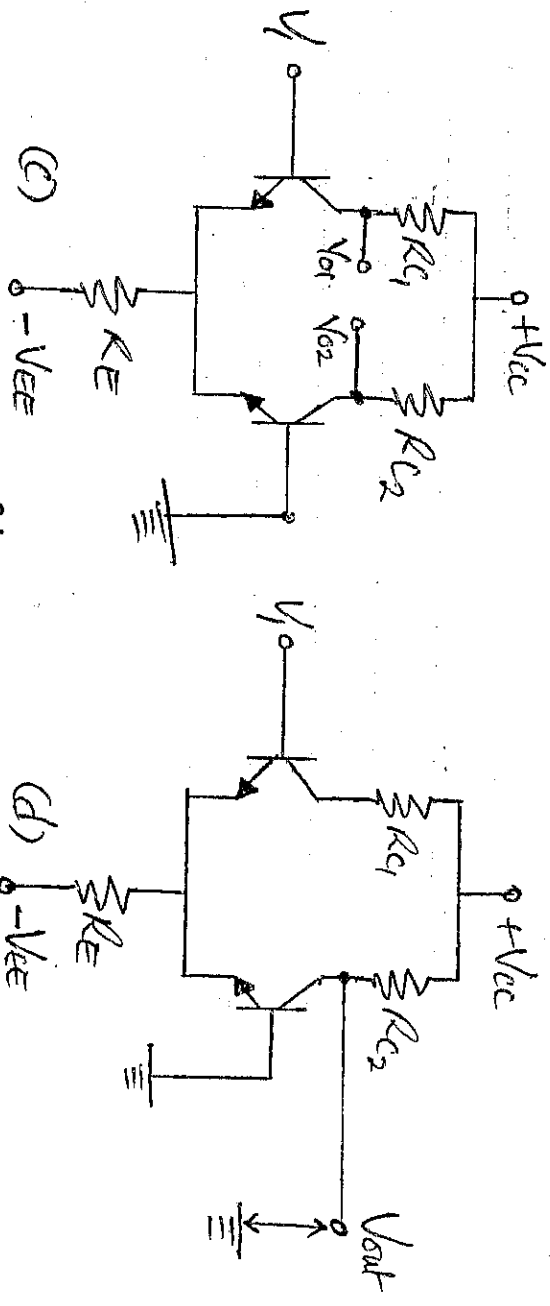
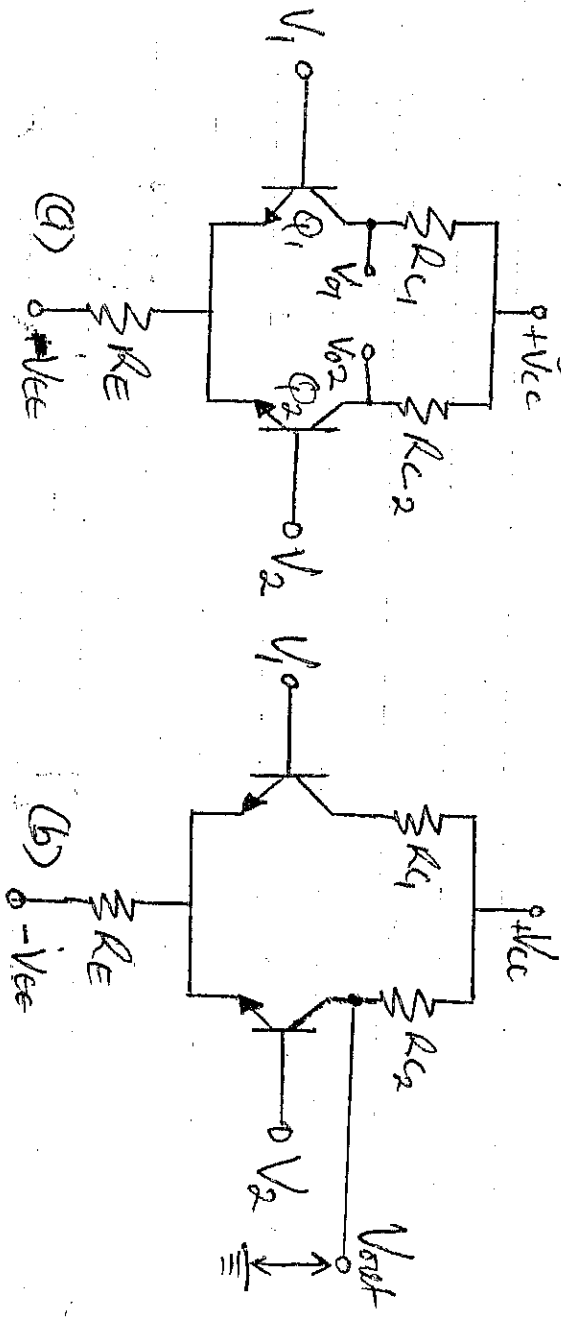


Fig 3.2

3.3

D.C Analysis:

To obtain the operating points for differential amplifier, the equivalent circuit is drawn by reducing all ac sources (i.e V_1 and V_2) to zero.

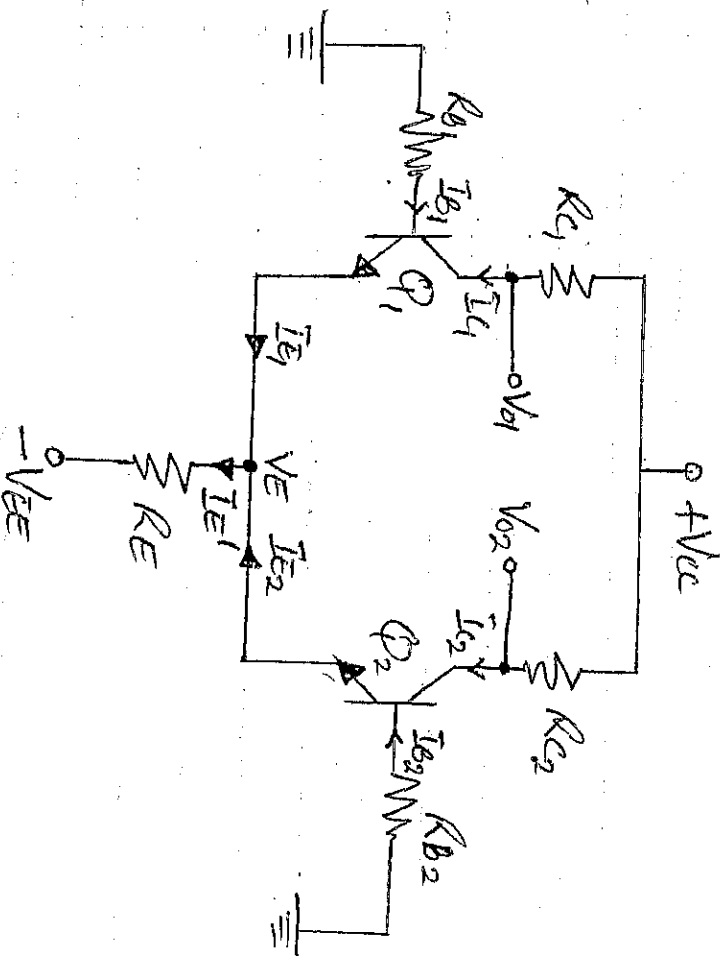


Fig 3.3: DC equivalent circuit.

From Fig 3.3,

$$V_E = -V_{EE}, \quad I_E' = I_{E1} + I_{E2} = 2I_E$$

$$\therefore I_E = \frac{V_E - (-V_{EE})}{2R_E + R_{B1}}$$

Since the circuit is balanced;

$$I_{E1} = I_{E2} = \frac{1}{2} I_E$$

$$I_{C1} = I_{C2} \approx \frac{1}{2} I_E, \quad I_{B1} = I_{B2} = \frac{I_E}{\beta} = \frac{I_{C1}}{\beta}$$

$$V_{01} = V_{CC} - I_{C1} R_{C1}$$

$$V_{02} = V_{CC} - I_{C2} R_{C2}$$

$$\text{Also } I_C \approx I_E = \frac{V_{EE} - V_{BE}}{2R_E}$$

if $\frac{R_B}{\beta} \ll 2R_E$ because

$$V_{EE} = V_{CC} + V_{BE} - I_C R_E$$

R_B is the internal resistance

of the source.

Example: The following specifications are given for the dual input, balanced output differential amplifier of Fig 3.3. $R_C = 2.2 \text{ k}\Omega$, $R_B = 50 \Omega$, $+V_{CC} = 10 \text{ V}$, $-V_{EE} = -10 \text{ V}$

$$R_E = 4.7k\Omega$$

$\beta = 100$, $V_{BE} = 0.715V$. Determine the operating points I_{CQ} and V_{CEQ} . Also find the balanced output.

Soln:

$$I_{CQ} = I_E = \frac{V_{EE} - V_{BE}}{2R_E + R_B/\beta} = \frac{10V - 0.715V}{2 \times 4.7k\Omega + \frac{50}{100}}$$

$$I_{CQ} = 0.988mA$$

$$\begin{aligned} V_{CEQ} = V_{CE} &= V_{CC} + V_{BE} - I_{CQ}R_C \\ &= 10 + 0.715 - (0.988mA \times 2.2k\Omega) \\ &= 10.715 - 2.1736 \\ &= 8.54V \end{aligned}$$

The output voltages $V_{O1} = V_{CE} - I_{CQ}R_C$

$$\begin{aligned} &= 10V - 2.1736V \\ &= 7.83V. \\ \therefore V_{O1} = V_{O2} &= 7.83V. \end{aligned}$$

3.4

AC Analysis:

To carryout the AC analysis, it is necessary to suppress all DC sources and draw the AC equivalent circuit of Fig 3.1(b) as shown in Fig 3.4.

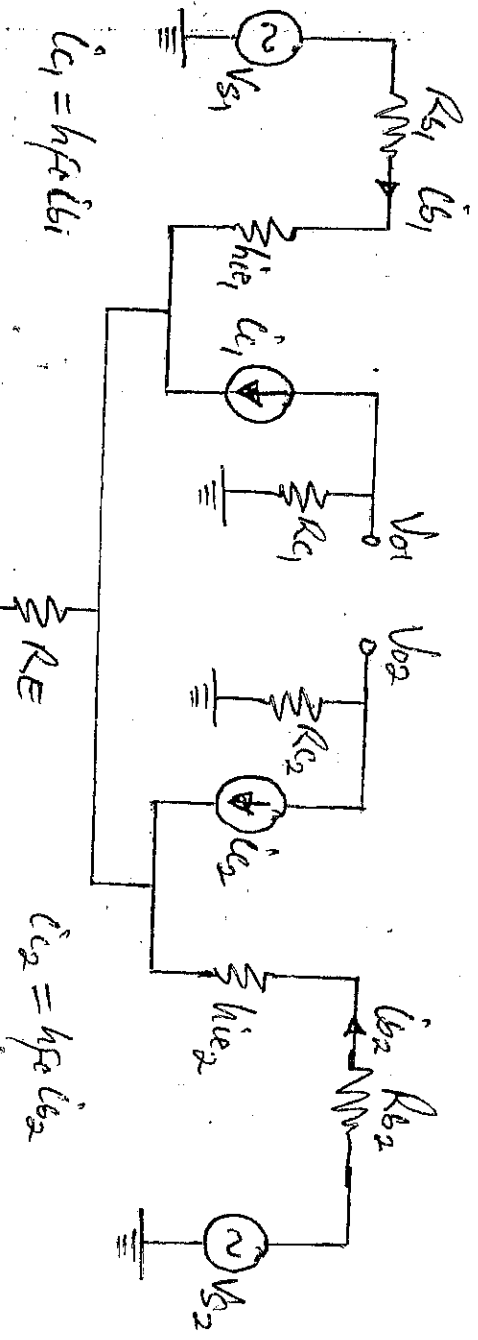


Fig 3.4: AC equivalent.

From Fig 3.4, the impedance of a current source is infinite.
Thus Fig 3.4 reduces to Fig 3.4.1.

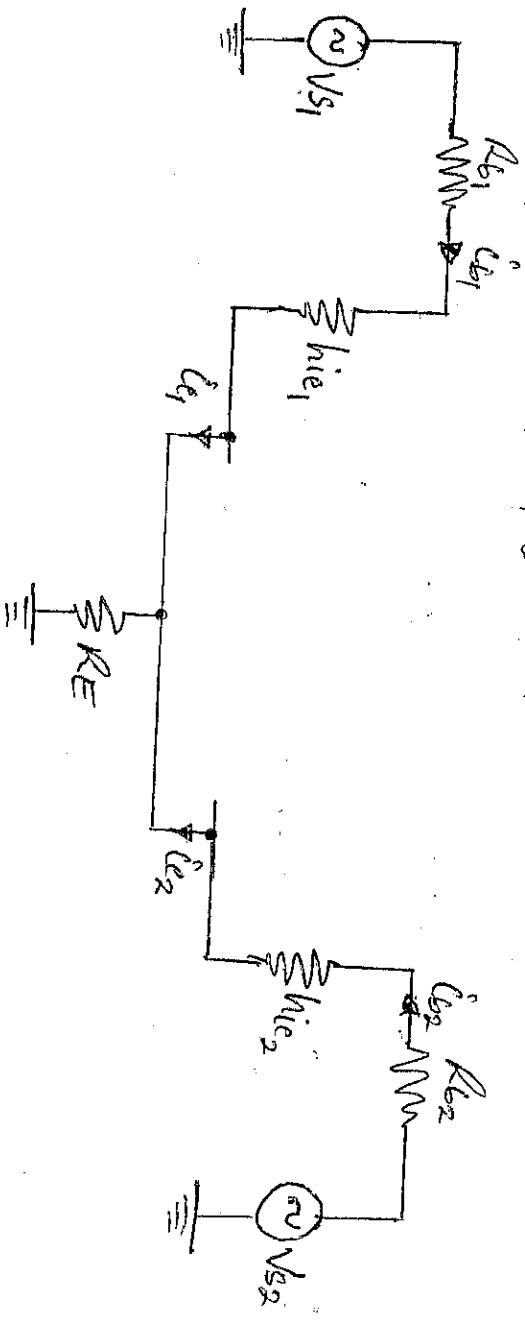
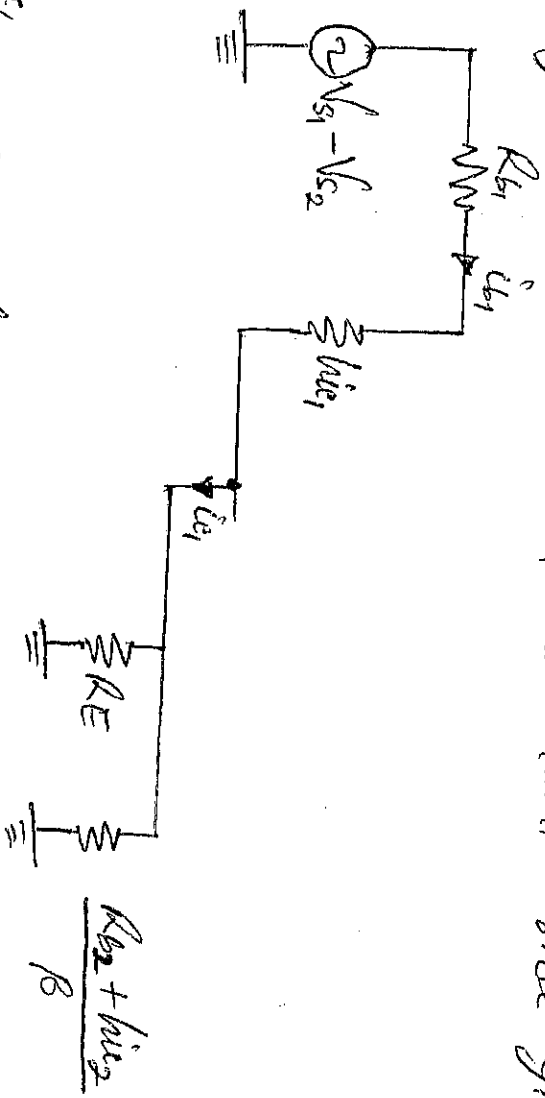
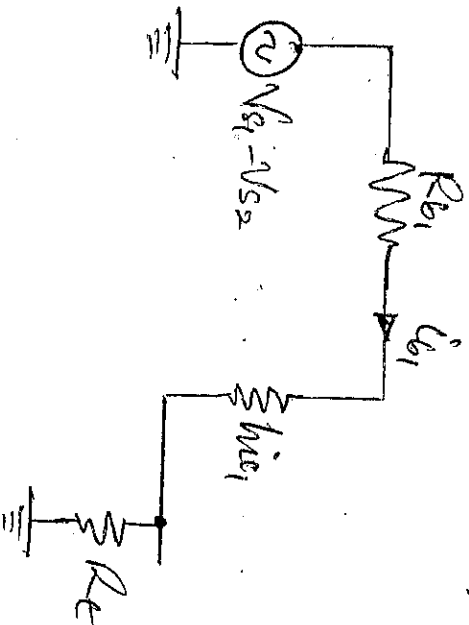


Fig 3.4.1: Equivalent circuit with collector open-circuited.

Referring the base network to the emitter side yields:

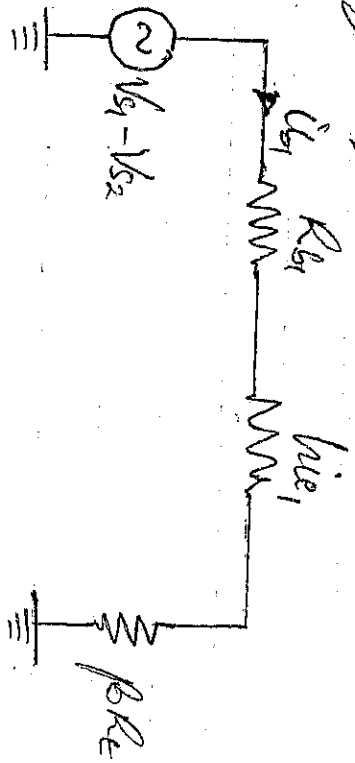


This network then becomes;



$$\text{where } R_t = R_E \parallel \left(\frac{R_{b2} + h_{ie2}}{\beta} \right)$$

And finally, referring the emitter side to the base side of Q_1 yields;



$$\therefore i_{b1} = \frac{V_{b1} - V_{s2}}{R_{b1} + h_{ie1} + \beta R_E}$$

$$\text{When } R_E = R_E \parallel \frac{R_{b2} + h_{ie2}}{\beta} = \frac{R_E (R_{b2} + h_{ie2})}{\beta R_E + R_{b2} + h_{ie2}}$$

If β is very large, then $\beta R_E \gg R_{b2} + h_{ie2}$

$$\therefore R_E = \frac{R_E (R_{b2} + h_{ie2})}{\beta R_E} = \frac{R_{b2} + h_{ie2}}{\beta}$$

Thus $\beta R_E = R_{b2} + h_{ie2}$, then

$$i_b = \frac{V_{b1} - V_{s2}}{R_{b1} + h_{ie1} + R_{b2} + h_{ie2}};$$

$V_{b1} - V_{s2}$ = difference voltage at the input = V_d

Since the circuit is symmetrical, $R_{b1} = R_{b2} = R_b$, $h_{ie1} = h_{ie2} = h_{ie}$

$$\therefore i_b = \frac{V_d}{2(R_b + h_{ie})}$$

$$V_{or} = -i_c R_{c1} = -h_{fe} i_{b1} R_{c1}$$

Substituting the value of i_b gives

$$V_{or} = \frac{-h_{fe} R_{c1}}{2(R_b + h_{ie})} \cdot V_d = \frac{-h_{fe} R_c}{2(R_b + h_{ie})} \cdot V_d$$

The differential gain A_d is defined as the ratio of output voltage to the difference input voltage.

$$\text{Thus; } A_d = \frac{V_o}{V_d} = \frac{-h_{fe} R_c}{2(R_b + h_{ie})}$$

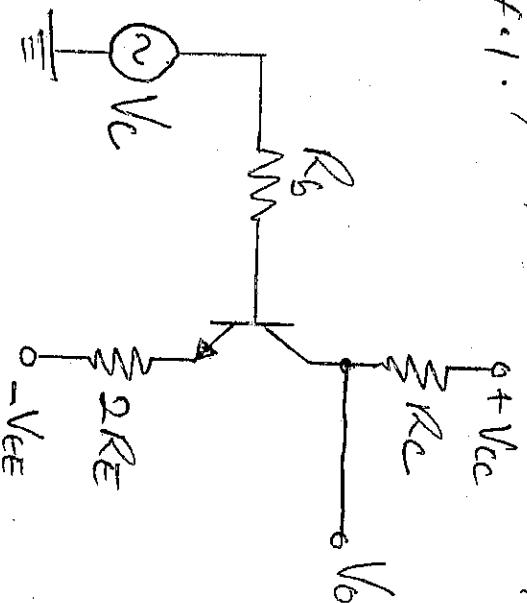
The input impedance seen by the source is;

$$R_{in} = R_{b1} + h_{ie1} + \beta R_c$$

$$= 2(R_b + h_{ie})$$

The output impedance $R_{out} = R_c$

For the common mode gain, we split the differential amplifier circuit of Fig 3.4.1 (b) into two equal parts, and analyse one half of the circuit. This yields the circuit of Fig 3.4.1.

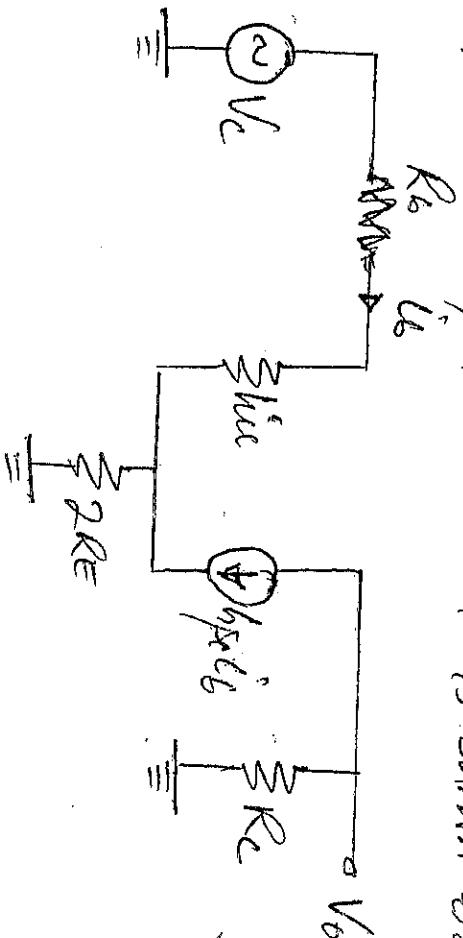


V_c = Common mode
input voltage

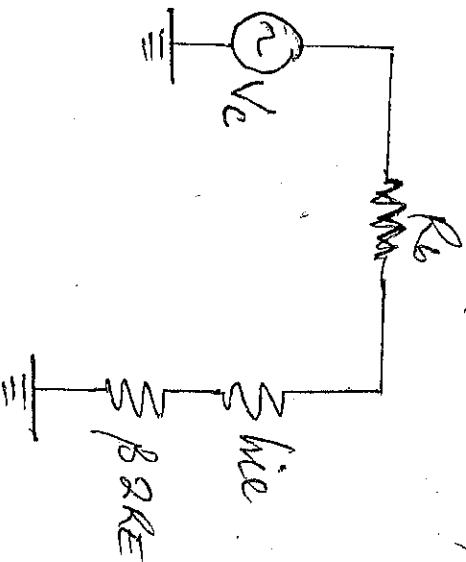
$$V_c = \frac{1}{2}(V_{s1} + V_{s2})$$

Fig 3.4.1: one-half of differential amplifier

The AC equivalent of this circuit is shown below;



The impedance of current source is infinite, thus the collector can be assumed to be open-circuited. Referring the emitter to the base yields;



$$\text{Thus } i_b = \frac{V_c}{R_b + h_{ie} + \beta R_E}$$

but $\beta R_E \gg R_b + h_{ie}$

$$\therefore i_b = \frac{V_c}{\beta R_E}$$

$$V_o = -i_c R_c = -\beta i_b R_c$$

substituting for i_b gives

$$V_o = -\frac{\beta V_c R_c}{\beta R_E} = -\frac{R_c \cdot V_c}{R_E}$$

The common mode gain $A_c = \frac{V_o}{V_c} = -\frac{R_c}{2R_E}$

The common mode rejection ratio (CMRR) of the differential amplifier is defined as the ratio of the differential gain to the common mode gain.

that is: $CMRR = \frac{A_d}{A_c} = \frac{-\frac{h_{fe} R_c}{2(R_b + h_{ie})}}{-\frac{R_c}{2R_E}}$

$$CMRR = \frac{h_{fe} R_E}{R_b + h_{ie}} \quad V_o = -V_d \frac{h_{fe} R_c}{2(R_b + h_{ie})}$$

In dB, CMRR is given as:

$$CMRR(dB) = 20 \log_{10} (CMRR)$$

$$A_d = \frac{V_o}{V_d} = -\frac{h_{fe} R_c}{2(R_b + h_{ie})}$$

Example: If the differential amplifier of Fig 3.1 (b) has the following parameters; $R_{b1} = R_{b2} = 21k\Omega$, $R_E = 20k\Omega$, $R_{b1} = R_{b2} = 1k\Omega$, $h_{ie1} = h_{ie2} = 14k\Omega$, $h_{fe1} = h_{fe2} = 40$. Determine the following: (i) A_d , (ii) A_c and (iii) CMRR in dB.

i) $A_d = \frac{-\frac{h_{fe} R_c}{2(R_b + h_{ie})}}{2(R_b + h_{ie})} = \frac{-40 \times 21,000\Omega}{2(1000\Omega + 14,500\Omega)}$

$$A_d = \frac{-840 \times 10^3 \Omega}{30 \times 10^3 \Omega} = -28$$

ii) $A_c = \frac{-R_c}{2R_E} = \frac{-21 \times 10^3 \Omega}{2 \times 20 \times 10^3 \Omega} = -0.525$

iii) $CMRR = \frac{h_{fe} R_E}{R_b + h_{ie}} = \frac{A_d}{A_c} = -28 / -0.525$

$$CMRR = 53.33$$

$$CMRR(dB) = 20 \log_{10} (53.33) \text{ dB}$$

$$CMRR(dB) = 34.54 \text{ dB}.$$

Assignment 1

1. The collector resistors of a differential amplifier are $8.2 \text{ k}\Omega$. The DC emitter current of each transistor is 0.89 mA . A 3 mV signal is applied to one input. Calculate the voltage gain, the single ended output voltage, and the differential output voltage.

2. What does common-mode rejection mean?

4.0 Introduction To OP-AMPS:

The term "operational amplifier" was first introduced by workers in the analogue computer field to denote an amplifier circuit that is capable of performing various mathematical operations such as subtraction, summation, differentiation, integration, and so on. But the term is now more loosely used to designate a direct-coupled high gain amplifier to which feedback may be added to confer either dB gain, frequency, or phase characteristics.

The operational amplifier offers several advantages which include small size, high reliability, low cost, temperature stability, short circuit protection, and low bias current requirement.

4.1 Basic Features of OP-AMPS:

Typically, OP-AMPS have 5-terminals as shown in

Fig 4.1(a) below:

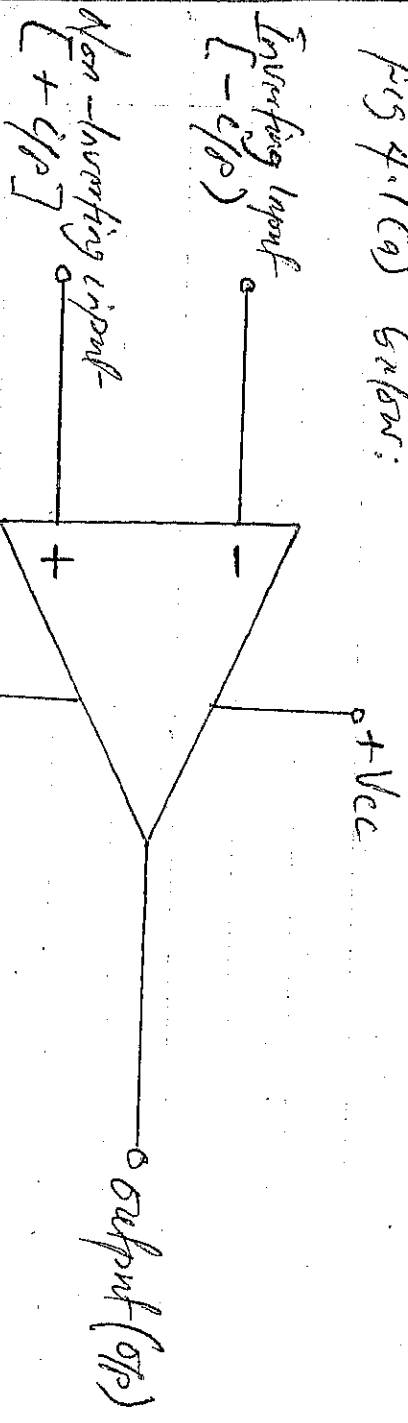


Fig 4.1(a): OP-Amp symbol.

The equivalent circuit is shown in Fig 4.1(b):

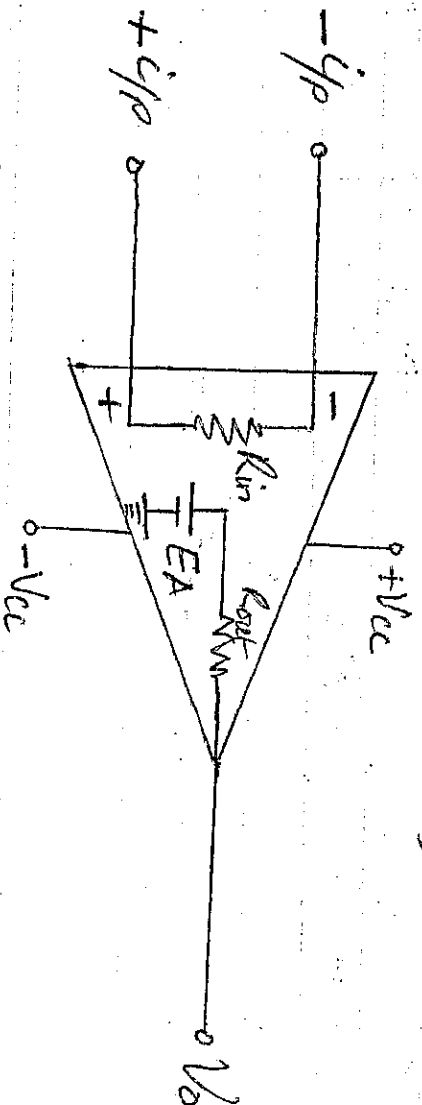


Fig 4.1(b): OP-Amp equivalent circuit-

R_{in} = input resistance	offset null	1		8	No conn.
R_{out} = output resistance	inverting	2		7	+V
A = open-loop gain	Non-inverting	3		6	output
E = Difference Voltage	-V	4		5	offset null

Standard IC packet.

Characteristics of OP-Amps:

1. OP-amps generally possess the following characteristics;
2. Very high input impedance in the order of Mega Ohms.
3. The input current to the op-amp is negligibly small.
4. Have very low output resistance.
5. The open-loop voltage gain is very high in the order of hundreds of thousands.
6. Have very wide bandwidth in the order of several

6. Margabats.
The two input terminals are at equipotential under bias condition.

4.3 Basic Concepts of OP-Amps:
Conceptually, the op-amp is made up of three basic building blocks viz; input stage, intermediate stage, and output stage as illustrated in Fig 4.2.

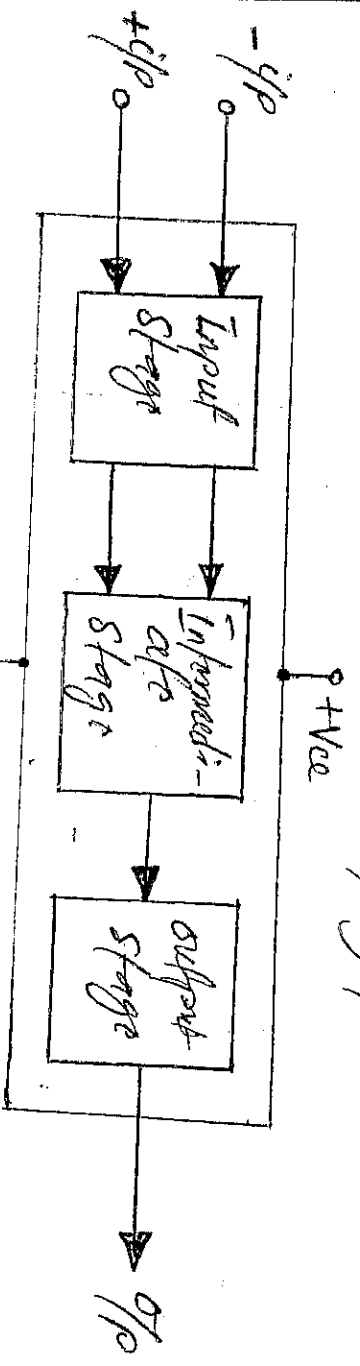
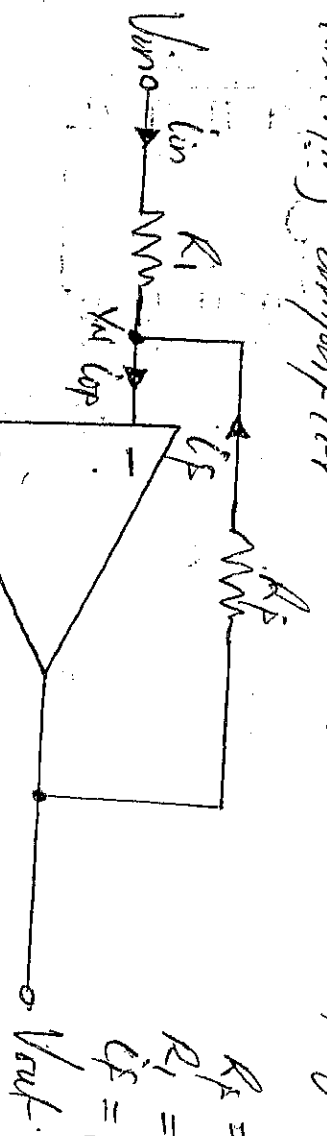


Fig 4.2: Internal building blocks of op-amps

4.3.1 Op-amp as an inverting amplifier.
Consider the op-amp of Fig 4.3.1 configured as an inverting amplifier



R_F = Feedback resistor
 R_1 = input resistor
 i_F = Feedback current
 i_{in} = input current
 i_{op} = op-amp current

Fig 4.3.1: Inverting Amplifier

From the diagram, using KCL at input node:
 $i_{in} + i_F = i_{op}$
but $i_{op} = 0$ Since input current is negligible.
 $\therefore i_{in} = -i_F$

$V_d = 0$ [virtual ground] terminals of op-amp are at equipotential.

$$\text{Now, } i_{in} = \frac{V_{in} - V_d}{R_i} = \frac{V_{in}}{R_i}$$

$$i_F = \frac{V_{out} - V_d}{R_F} = \frac{V_{out}}{R_F}$$

$$\text{Thus, } \frac{V_{in}}{R_i} = -\frac{V_{out}}{R_F}$$

The amplifier voltage gain is thus: $A_v = \frac{V_{out}}{V_{in}}$

$$\Rightarrow A_v = \frac{V_{out}}{V_{in}} = -\frac{R_F}{R_i}$$

$$\text{The input impedance } Z_{in} = \frac{V_{in}}{i_{in}} = \frac{i_{in} R_i}{i_{in}} = R_i$$

$$\text{The output impedance } Z_{out} = \frac{V_{out}}{i_{out}} = \frac{V_{out}}{i_F} = \frac{i_F R_F}{i_F} = R_F$$

4.3.2 Op-amp as a non-inverting amplifier.

Here the input signal is fed to the non-inverting terminal of the op-amp as illustrated in Fig 4.3.2.

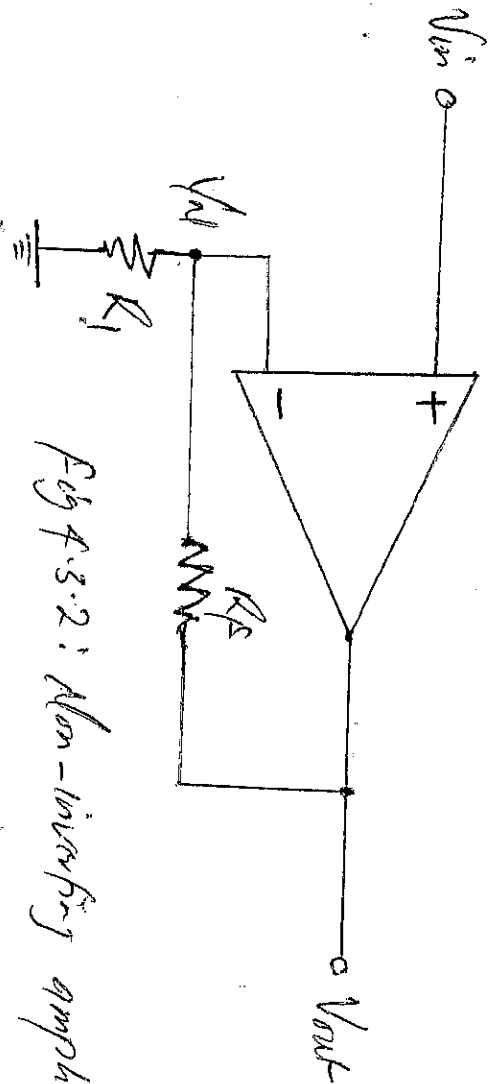


Fig 4.3.2: Non-inverting amplifier

From the circuit above, $V_M = V_{in}$

$$\text{Thus } V_M = \frac{R_1}{R_1 + R_F} \cdot V_{out} = V_{in}$$

$$\therefore A_v = \frac{V_{out}}{V_{in}} = \frac{R_1 + R_F}{R_1} = \left(1 + \frac{R_F}{R_1}\right)$$

Thus the non-inverting amplifier has a higher amplification than the inverting configuration.

$$\text{The input impedance } Z_{in} = \frac{V_{in}}{i_{in}}$$

$$\text{but } i_{in} = i_{op} = 0$$

$$\text{Thus } Z_{in} = \frac{V_{in}}{0} = \infty \text{ [high input impedance]}$$

$$Z_{out} = \frac{V_{out}}{i_F}$$

$$i_F = \frac{V_{out}}{R_1 + R_F}$$

$$\therefore Z_{out} = \frac{V_{out}}{V_{out}} \cdot R_1 + R_F = R_1 + R_F$$

A3.3 Op-amp as a buffer (Voltage follower)

The op-amp voltage follower or buffer amplifier as it is mostly called is configured as shown below:

$$\text{Here } V_M = V_{out} = V_{in}$$

$$\text{Thus } A_v = \frac{V_{out}}{V_{in}} = 1$$

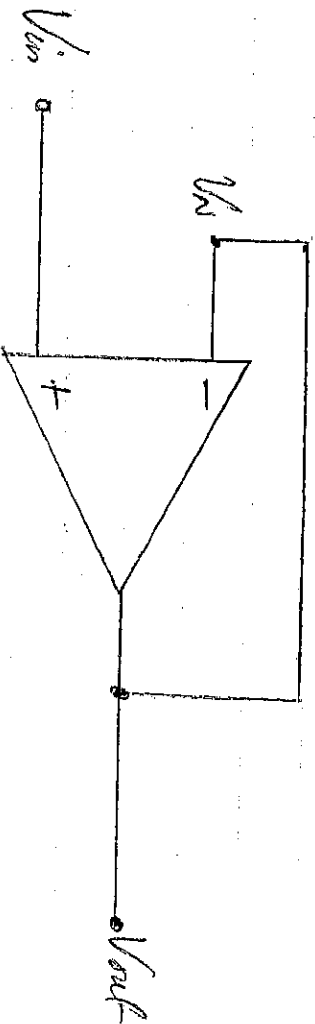


Fig 4.3.3: Buffer amplifier configuration.

4.3.4 Op-amp as a Summing Amplifier.

The summer could be in the inverting or non-inverting mode. The basic structure of the inverting summer is shown in Fig 4.3.4(a).

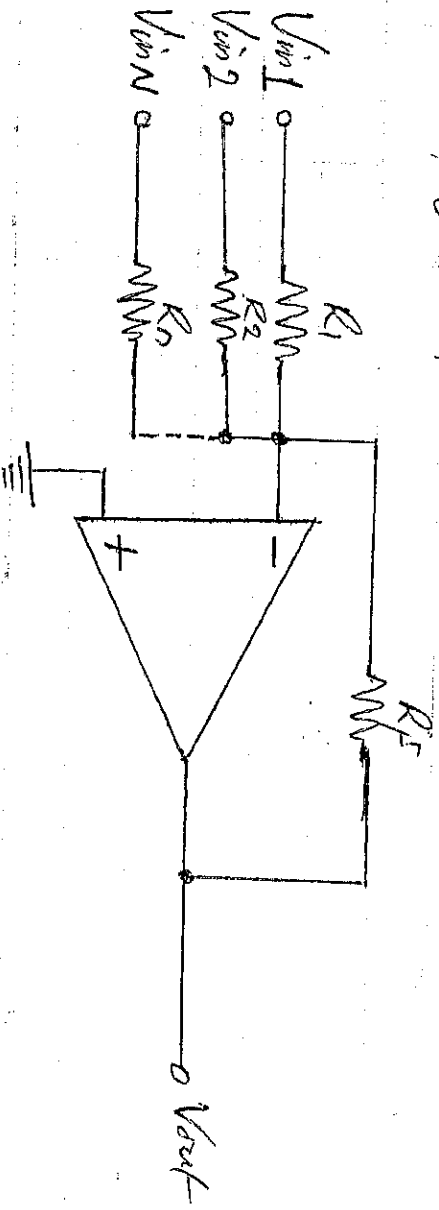


Fig 4.3.4(a): Inverting Summer.

To analyse this circuit, we apply superposition theorem.

Let V_{out}' be Voltage due to V_{in1} .

V_{out}'' be Voltage due to V_{in2}

V_{out}''' be Voltage due to V_{inN}

Thus $V_{out}' = -\frac{R_f}{R_1} V_{in1}$

$$V_{out}'' = -\frac{R_f}{R_2} V_{in2}$$

$$V_{out}'' = -\frac{R_F}{R_N} V_{inN}$$

Superposing these output yields the overall output as:-

$$\begin{aligned} V_{out} &= V_{out}' + V_{out}'' + V_{out}''' \\ &= -\left\{ \frac{R_F}{R_1} V_{in1} + \frac{R_F}{R_2} V_{in2} + \frac{R_F}{R_N} V_{inN} \right\} \\ &= -R_F \left\{ \frac{V_{in1}}{R_1} + \frac{V_{in2}}{R_2} + \frac{V_{inN}}{R_N} \right\} \end{aligned}$$

$$\therefore V_{out} = -R_F \sum_{i=1}^N \frac{V_{in i}}{R_i}$$

In the non-inverting mode, the summer could be configured as shown in Fig 4.3.4 (b).

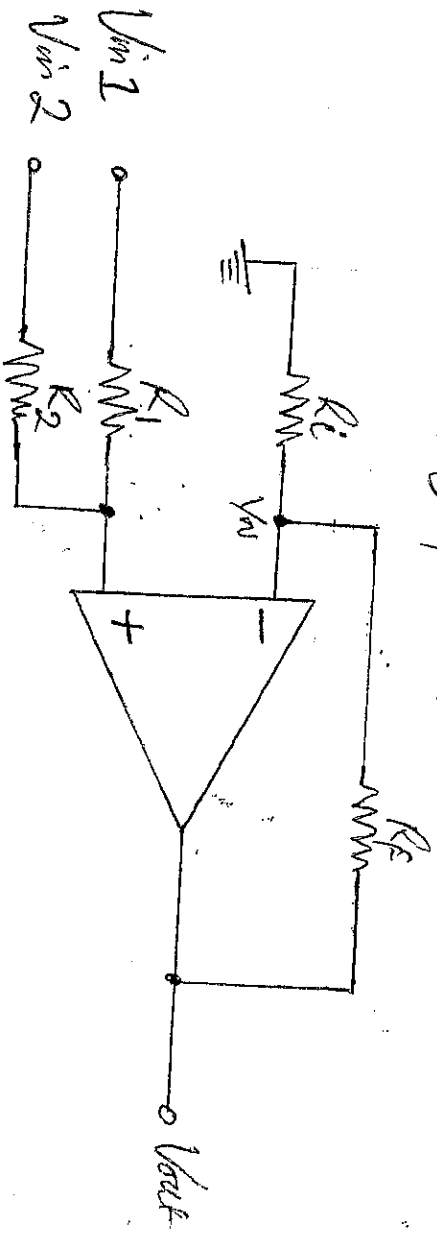


Fig 4.3.4 (b): Non-inverting summer

$$V_{out} = \left(1 + \frac{R_F}{R_i}\right) V_M$$

due to V_{in1} ; $V_M = \left(\frac{R_2}{R_1 + R_2}\right) V_{in1}$

$$\therefore V_{out}' = \left(1 + \frac{R_F}{R_i}\right) \times \left(\frac{R_2}{R_1 + R_2}\right) V_{in 1}$$

due to $V_{in 2}$; $V_{in} = \left(\frac{R_1}{R_1 + R_2}\right) V_{in 2}$

$$\therefore V_{out}'' = \left(1 + \frac{R_F}{R_i}\right) \left(\frac{R_1}{R_1 + R_2}\right) V_{in 2}$$

$$\Rightarrow V_{out} = V_{out}' + V_{out}''$$

$$V_{out} = \left(1 + \frac{R_F}{R_i}\right) \left(\frac{R_2}{R_1 + R_2}\right) V_{in 1} + \left(1 + \frac{R_F}{R_i}\right) \left(\frac{R_1}{R_1 + R_2}\right) V_{in 2}$$

$$V_{out} = \left\{ \left(1 + \frac{R_F}{R_i}\right) \right\} (R_2 V_{in 1} + R_1 V_{in 2})$$

Summing amplifiers find extreme applications in audio mixer. Also, summers could be used to perform digital to analogue conversions.

Example: If the inverting summer of Fig 4.3.4(a) has: $R_F = 10k\Omega$, $R_1 = 10k\Omega$, & $R_2 = 20k\Omega$, determine the output for the following inputs;

(a) $V_{in 1} = +1V$, $V_{in 2} = -1V$

(b) $V_{in 1} = +4V$, $V_{in 2} = -3V$

Solution: For inverting Amp $A_v = \frac{V_{out}}{V_{in}} = -\frac{R_F}{R_i} \Rightarrow V_{out} = -\frac{R_F}{R_i} \times V_{in}$

(a)

For $V_{in 1} = +1V$; $V_{out 1} = -\frac{R_F}{R_1} \times V_{in 1} = -\frac{10k\Omega}{10k\Omega} \times 1V$

$$V_{out} = -1V$$

For $V_{in2} = -1V$; $V_{out} = -\frac{R_F}{R_2} \times V_{in2} = -\frac{10k\Omega}{20k\Omega} \times (-1V)$

$V_{out} = +500mV$

(6)

For $V_{in1} = +4V$; $V_{out} = -\frac{R_F}{R_1} \cdot V_{in1} = -\frac{10k\Omega}{10k\Omega} \times 4V$

$V_{out} = -4V$

For $V_{in2} = -3V$; $V_{out} = -\frac{R_F}{R_2} \cdot V_{in2} = -\frac{10k\Omega}{20k\Omega} \times (-3V)$

$V_{out} = 1.5V$

Example: Given a basic two inputs non-inverting summer with $R_i = 1k\Omega$, $R_F = 9k\Omega$, $R_1 = 2k\Omega$, and $R_2 = 3k\Omega$. Determine the output for the following inputs:

(a) $V_{in1} = +2V$, $V_{in2} = -2V$

(b) Also for the total output.

Solution:

(a)

For $V_{in1} = +2V$, $V_{out1} = \left(1 + \frac{R_F}{R_i}\right) \left(\frac{R_2}{R_1 + R_2}\right) \cdot V_{in1}$

$V_{out1} = \left(1 + \frac{9}{1}\right) \left(\frac{3}{2+3}\right) \times 2V = 10 \times 0.6 \times 2V$

$V_{out1} = 12V$.

For $V_{in2} = -2V$, $V_{out2} = \left(1 + \frac{R_F}{R_i}\right) \left(\frac{R_1}{R_1 + R_2}\right) \cdot V_{in2}$

$V_{out2} = \left(1 + \frac{9}{1}\right) \left(\frac{2}{2+3}\right) \times (-2V) = 10 \times 0.4 \times (-2V)$

$V_{out2} = -8V$

(6) $V_{out\ total} = 12V + (-8V)$

$V_{out\ total} = 12V - 8V = \underline{\underline{4V}}$

4.3.5 OP-amp as a differentiator

As a differentiator, the op-amp is configured as shown in Fig 4.3.5.

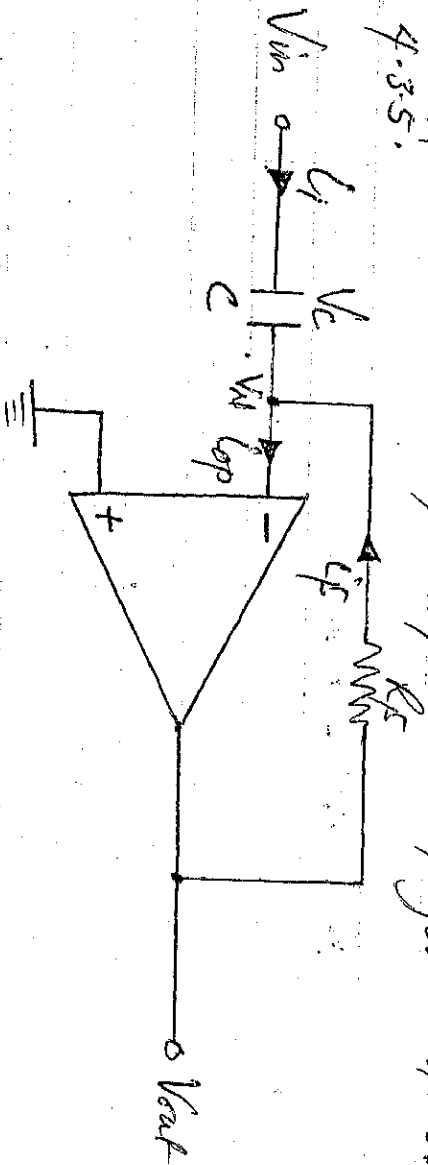


Fig 4.3.5: OP-amp as differentiator

$\therefore i_i + i_{cf} = i_{op} = 0$

$i_i = -i_{cf}$

$i_{cf} = \frac{V_{out}}{R_f}$, $i_i = \frac{dq}{dt}$; $q = \text{charge across capacitor}$

but $q = CV_c$; $V_c = V_{effg}$ across capacitor.

$i_i = \frac{d[CV_c]}{dt} = C \frac{dV_{in}}{dt}$

Now $C \frac{dV_{in}}{dt} = -\frac{V_{out}}{R_f}$

$V_{out} = -C R_f \frac{dV_{in}}{dt}$

Thus the output voltage is proportional to the derivative of the input voltage.

Q3.6

Op-amp as Integrator.

The configuration of the integrator is shown below;

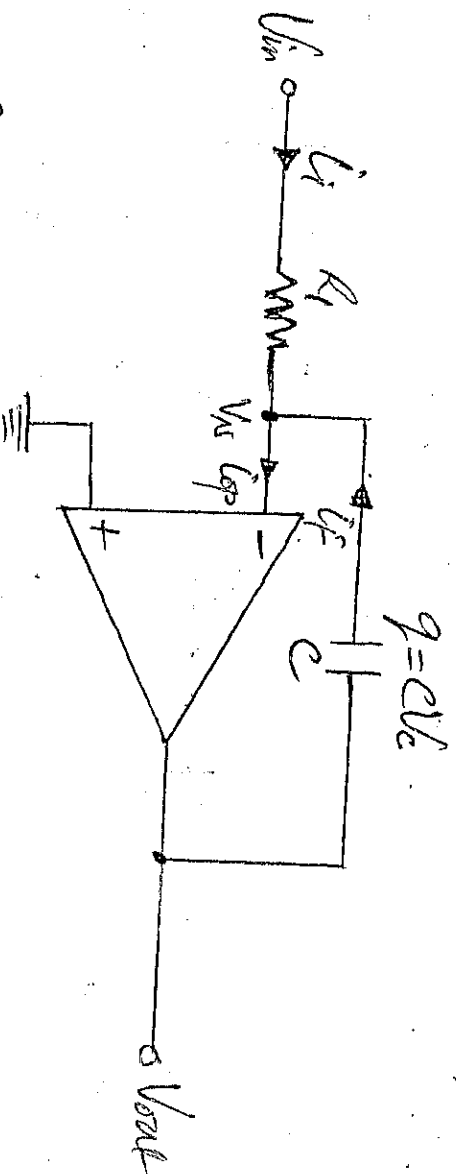


Fig 4.3.6: Op-amp as integrator.

$$i_f = \frac{dq}{dt} = C \frac{dV_c}{dt} = C \frac{dV_{out}}{dt}$$

$$i_i = \frac{V_{in}}{R_1}$$

$$i_i = -i_f \quad ; \quad \frac{V_{in}}{R_1} = -C \frac{dV_{out}}{dt}$$

$$\frac{dV_{out}}{dt} = -\frac{1}{CR_1} \cdot V_{in}$$

$$dV_{out} = -\frac{1}{CR_1} V_{in} dt \quad \text{integrating both sides gives;}$$

$$V_{out} = -\frac{1}{CR_1} \int V_{in} dt$$

Thus the output voltage is proportional to the integral of

The input signal. both the differentiator and the integrator find application in process control (such as PID's).

Example: Given an integrator circuit of Fig 4.36, its single square wave pulses of 50ms duration is applied to an input resistance of 4.7k Ω and the feedback capacitor is 0.5 μ F, what is the maximum output voltage.

Solution:

$$V_{out} = -\frac{1}{CR_1} \int_0^t V_{in} dt.$$

$$= -\frac{1}{CR_1} \left[V_{in} t \right]_0^t$$

$$= -\frac{1}{0.5 \times 10^{-6} \times 4.7 \times 10^3} \left[200 \times 10^{-3} \times 0.05 \right]$$

$$= -425.5 (0.01)$$

$$V_{out} = -4.255V.$$

4.4 Other Areas of Application of Op-amps.

4.4.1 Voltage Limiter:

Voltage limiters or clippers as they are sometimes called, are circuits designed to limit the amplitude of voltage that is passed on to a load. The limiter guarantees that the voltage applied to the load device will not pass (or exceed) a certain predetermined level. They are very useful for any application that requires over-voltage protection. Fig 4.4.1 shows Zener diode voltage limiter using op-amp 741.

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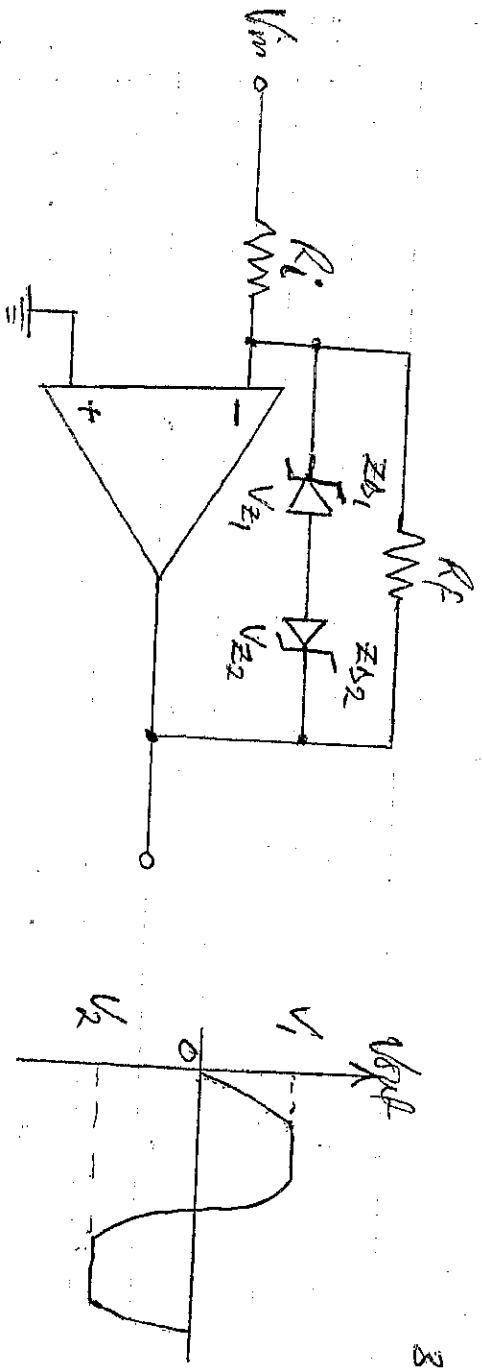


Fig 4.4.1: Voltage limiter

During the negative saturation;

When $V_{out} \geq 0.7V$;

ZD_2 is reverse biased and the voltage drop across it is equal to V_{Z2} , ZD_1 is forward biased and the voltage drop across it is $0.7V$.

$$\therefore V_{out(max)} = 0.7V + V_{Z2} = V_1$$

During the positive saturation; when $V_{out} \leq -0.7V$, ZD_1 is reverse biased and the voltage drop across it is equal to V_{Z1} , ZD_2 is forward biased and the voltage drop across it is equal to $-0.7V$.

$$\therefore V_{out(min)} = -0.7 - V_{Z1} = -(0.7 + V_{Z1}) = -V_2.$$

When $V_{Z1} = V_{Z2}$, the resulting clipper is termed symmetrical limiter. If however $V_{Z1} \neq V_{Z2}$, the circuit is termed asymmetrical. The minimum voltage at which clipping begins is given by:

$$V_{in(min)} = \frac{V_{out(clipper)}}{A_v}$$

Oscillator:

An oscillator is an electronic circuit designed to continuously generate a periodic waveform of a particular frequency. It converts DC into some form of pulsating AC or AC. Oscillators find application in many systems which include: Radio & television, transmitters & receivers, UPS, microwave, computers etc.

Fig 4.4.2 shows a Wien-Bridge oscillator, which is a typical oscillator exhibiting the 0° phase shift. This is because the feedback network is connected to the non-inverting input of the op-amp as illustrated.

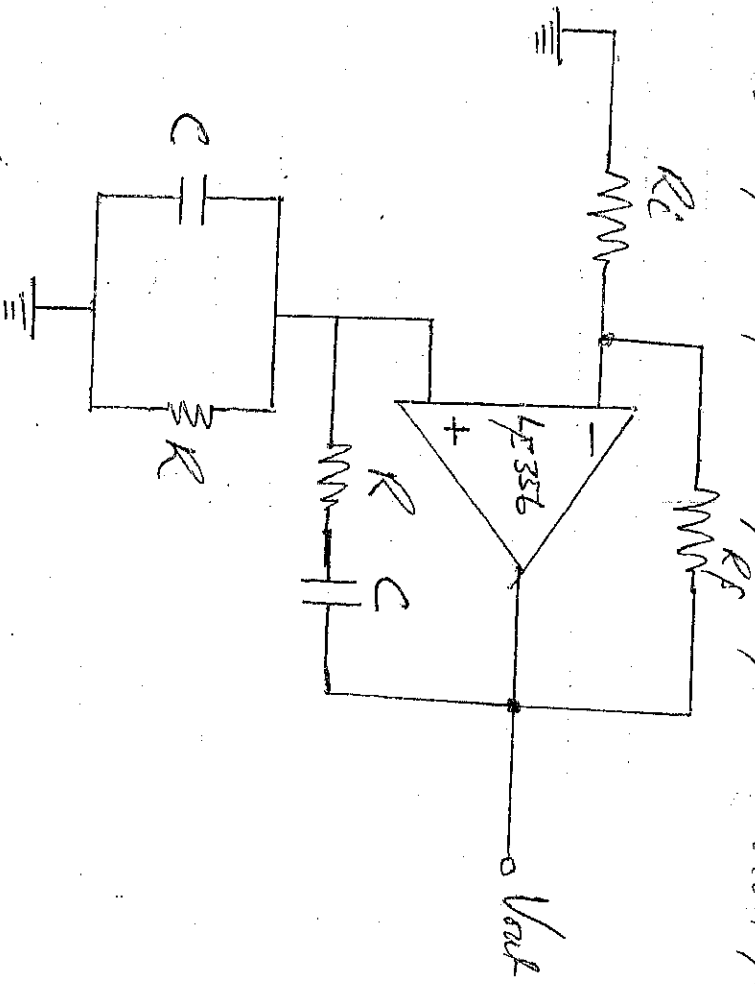


Fig 4.4.2: Wien Bridge Oscillator.

Voltage Comparators:

A "comparator" is an electric circuit that compares two input voltages. One voltage is called the reference voltage (V_{ref}) and the other is called the input voltage (V_{in}). When the input is above or falls below V_{ref} , the output changes polarity from positive to negative or vice versa.